Vignesh S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.44
Sree Narayana Public School, Vadakkevila, Kollam, Kerala	Class XII (CBSE)	2019	91.60%
Ideal Indian School, Doha, Qatar	Class X (CBSE)	2017	10.00%

PUBLICATIONS

FaultSense: Fault Localization in Large-Scale Mixture-of-Experts Model Serving Infrastructure

*Harish S A, **Vignesh S,** Ashwin Kurella, Rohan Gandhi, Praveen Tammana*

- Application-layer fault localization for large-scale MoE serving infrastructure.

17th ACM SIGOPS Asia-Pacific Workshop on Systems, Bangkok, Thailand

***DOI:** 10.1145/3838177.3841737*

Detecting Adversarial State Manipulation in In-Network Fast ReRoute Systems

*Harish S A, **Vignesh S,** Divya Pathak, Anil Kumar Sharma, Praveen Tammana*

- SmartNIC-based detection of adversarial manipulation in fast reroute systems

IEEE Transactions on Network and Service Management 2026

***DOI:** 10.1109/TNSM.2026.3715353*

ListGuard: Mitigating Pollution Attacks on In-Network Invertible Bloom Lookup Tables

*Harish S A, **Vignesh S,** Sathwik Kodamarthi, Bikraj Shrestha, Gollapudi Sasank, Mahanth Kumar Valluri, Pravein G Kannan, Praveen Tammana*

- Sampling-based detection and recovery from pollution attacks on in-network probabilistic data structures.

25th International Federation for Information Processing Networking Conference, Lugano, Switzerland

***DOI:** 10.23919/IFIPNetworking70592.2026.11579227*

CBPF: Efficient Packet Filtering Algorithm using Cascading Bloom Filter in Software Defined Networks

***Vignesh S,** Akhil M Reji, Erukala Charan Teja, Harish S A, Vidya Sagar Thalapala, Koppala Guravaiah*

- Designed a P4-based adaptive data-plane state management system using Cascading Bloom Filters.

2025 IEEE Future Networks World Forum (FNWF), Bengaluru, India

***DOI:** 10.1109/FNWF66845.2025.11317382*

WORK EXPERIENCE

Research Assistant

IIT Hyderabad

- Researching fault localization and reliability in large-scale LLM training and serving infrastructure
- Designed lightweight, probing-based diagnostic frameworks to localize faults in AI serving systems

- Implemented anomaly detection for fast control loop systems on SmartNICs under resource limits
- Evaluated proposed techniques across distributed AI workloads under diverse injected faults achieving >90% accuracy.

Intern - Project Associate

Mondee Tech Pvt. Ltd.

- Built an agentic AI framework using LangGraph to orchestrate multi-agent workflows for a travel app
- Implemented task coordination, tool invocation, and state management across cooperating agents
- Validated system behavior against realistic user queries, ensuring reliable end-to-end task completion
- Orchestrated 8 agents with LangGraph, Neo4j, and Pinecone for contextual task execution resolving queries within 10s.

Research Intern

IIT Hyderabad

- Analyzed robustness mechanisms for data-plane measurement systems under adversarial pollution.
- Prototyped probabilistic detection and recovery techniques for in-network monitoring pipelines.
- Validated the mechanisms on programmable SmartNIC platforms in a production-style deployment.

Cyber Forensics Intern

Centre for Development of Advanced Computing (C-DAC)

- Investigated the ComRAT v4.0 malware family through hands-on threat handling and behavioral analysis
- Analyzed malware samples to identify infection vectors, persistence mechanisms, and C2 communication patterns
- Gained practical exposure to malware detection and cyber threat handling workflows used in forensic investigations

PROJECTS

Failure Localization on Large-scale LLM serving Infrastructure

*Project work done in collaboration with **Microsoft Research India**.*

- Developed an application layer framework for fault localization in LLM-serving clusters for unprivileged tenants.
- Designed a scalable probing mechanism achieving 20X fewer probes than naive probing mechanisms
- Repurposed MoE model to be a lightweight probe execution mechanism using < 1% VRAM to execute probes
- Localized all injected faults in simulations up to 325 GPUs and probe traffic showed 1% inflation on production serving.

Low-Overhead Fail-Slow Fault Diagnostics for Large-Scale Distributed Training

- Designed fault-localization system using provenance graphs, achieving ~90% accuracy in concurrent fault scenarios
- Separated faulty components from propagated symptom via lag-gated explaining-away within 3-4 iteration from onset
- Built a per-node anomaly detector using a modified CUSUM algorithm, reducing diagnosis latency to under 24 seconds
- Engineered component-level tracing using lightweight metric buffering to reduce runtime monitoring overhead to <0.5%

Modular BGP Control Plane Verification via Local Invariant Checks

- Reproduced LightYear[SIGCOMM'23] a Python/Z3 verifier splitting BGP safety properties into per-edge checks.
- Covered no-transit, bogon, RTBH, isolation, and RPKI policies across 17 simulated test networks.
- Extended with CEGAR-based invariant inference and incremental re-verification of affected checks only.
- Verified 14-router, 186-edge networks (566 checks) in ~3.3 s with linear scaling with localized injected faults.

SmartNIC-based Adversial Manipulation Detection for In-Network Fast Control Loop Systems

Project work done with **National Security Council Secreteriat**

- Implemented Blink FRR system logic on a Netronome smartNIC (NFP-4000) using micro-C for line-rate processing
- Extended the data-plane pipeline to extract per-flow features feeding a control-plane anomaly detector
- Redesigned the pipeline to run anomaly detection entirely in the data plane, cutting detection latency by 99.95%
- Developed ^{2}-based anomaly detection on Blink, detecting attacks before reroute with 0.3% FPR and 0% FNR.

Mitigating Pollution Attacks on Data Plane-based Probabilistic Data Structures

Project in collaboration with **National Security Council Secreteriat**

- Designed a probabilistic sampling framework using sampled flows to recover statistics from pollution attacks on IBLTs
- Implemented the detection and recovery modules on a Netronome smartNIC using micro-C for line-rate processing
- Achieved full recovery at just 0.34-4% sampling overhead, with 2.83× lower memory cost than naive over-provisioning
- Introduced only 0.52% average latency and 0.026% throughput overhead, with 230× lower memory than LFFZ.

Efficient Packet Filtering in SDN Data Planes using Cascading Bloom Filters

B.Tech Thesis Project

- Designed a Cascading Bloom Filter for stateful session filtering, replacing cuckoo hashing in P4
- Implemented FIFO, LRTS, and packet-count eviction policies using P4 register read/write operations
- Achieved 0% illegitimate packets passed and ~0.0002% legitimate packet loss on 550K-packet traces
- Validated the design in Mininet and P4 for programmable data-plane deployment

SKILLS

Languages: C, C++, Python, Bash, SQL, P4, Micro-C, Golang (basic)

AI/ML: PyTorch, Deep Learning, LLM Serving (vLLM), Distributed Training (DDP), LangGraph, Horovod, DeepSpeed

Systems & Networking: Linux, Mininet, gRPC, Parallel Programming, Socket Programming, Data-plane Programming

Cloud & DevOps: Docker, Kubernetes, Istio, Google Cloud Platform (GCP), Prometheus, Grafana, Git, CI/CD, REST APIs

Data & Analysis: NumPy, Pandas, Matplotlib, Probabilistic Data Structures, Anomaly Detection

RELEVANT COURSES

Core Systems: Computer Organization & Architecture, Operating Systems, Computer Networks, Compiler Design

Advanced Systems: Topics in Networks, Parallel & Distributed Computing, Hardware Architecture for Deep Learning

Algorithms: Theory of Computation, Advanced Data Structures, Software Verification, Linear Optimization

AI/ML: Foundations of Machine Learning, Deep Learning, Soft computing, Applied Predictive Analysis

POSITIONS OF RESPONSIBILITY

Teaching Assistant

IIT Hyderabad

- Assisted 3 institute courses (~80 students): Computer Networks, Topics in Networks, and Distributed AI Training
- Conducted tutorials, graded assignments, ran lab sessions on wide variety of subjects

Secretary, Students' Co-operative Mess

IIIT Kottayam

- Led the post-COVID restart of a 1,500-student mess, coordinating a staff of 30 and 5 vendors for daily operations
- Managed a 1.5Cr budget and handled student grievances, resolving issues within 2 days

Lead, Photography Club

IIIT Kottayam

- Coordinated a team of 20 students for photography activities and media coverage of institute events.

Volunteer, *International Conferences*

- Volunteered for ACM India Event 2025, ICDCN 2024, ICITIIT 2023 and other small scale conferences.

- Supported event logistics including venue setup, local tour coordination.
- Assisted attendees and invited guests throughout the conferences with an average >100 participants

HOBBIES & INTERESTS

Sports & Games: Badminton, Cricket

Travel & outdoors: road trips, trekking, hiking

Games & Interests: Chess, E-football, Formula 1

Others: Photography, Driving (HMV license holder), Cooking